

Internet of Things (IoT) Training Program

NTI Smart Parking Management System

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Abstract

NTI Smart Parking Management System is an Internet of Things (IoT) based solution designed to automate parking management by integrating embedded hardware with cloud-based services. The system monitors parking slot availability in real time, controls the entrance gate using a servo motor, updates parking status through MQTT communication, and provides users with live parking information via a Node-RED dashboard.

The ESP32 microcontroller serves as the edge device responsible for monitoring parking events, controlling hardware components, and publishing parking data to the HiveMQ MQTT broker. Node-RED acts as the middleware platform that processes incoming data, updates the dashboard, calculates parking fees based on parking duration, and sends parking tickets and receipts through a Telegram Bot.

The proposed architecture follows the **Edge Computing + IoT Gateway** model, reducing the computational load on the ESP32 while enabling scalable and real-time communication. This approach provides an efficient, low-cost, and flexible smart parking solution suitable for educational and small-scale parking environments.

Introduction

With the rapid growth of the Internet of Things (IoT), traditional parking systems are being replaced by intelligent solutions that improve efficiency, reduce human intervention, and provide a better user experience. Smart parking systems help drivers quickly identify available parking spaces while enabling parking operators to monitor parking status in real time.

The NTI Smart Parking Management System is an IoT-based project that automates parking management using an ESP32 microcontroller, MQTT communication, Node-RED, and a Telegram Bot. The system monitors parking slot occupancy, controls the parking gate using a servo motor, displays parking information on an LCD screen, and updates users through a real-time dashboard.

Instead of hosting a web server on the ESP32, which requires significant memory and processing resources, the proposed system adopts an Edge Computing + IoT Gateway architecture. The ESP32 performs hardware control and publishes parking data via MQTT to the HiveMQ broker, while Node-RED handles data visualization, billing calculations, and Telegram notifications.

This architecture improves system scalability, reduces the computational load on the microcontroller, and enables efficient real-time communication between all system components.

Hardware Components

Component	Quantity	Purpose
ESP32 Development Board	1	Main microcontroller
Servo Motor (SG90)	1	Controls the parking gate
16×2 LCD with I2C Module	1	Displays parking status
Green LED	1	Indicates parking availability
Red LED	1	Indicates parking full status
Push Buttons	2	Simulate vehicle entry and exit
Resistors (220 Ω)	2	LED current limiting
Jumper Wires	As needed	Hardware connections

Software Tools

Software	Purpose
Wokwi Simulator	Hardware simulation
MicroPython	ESP32 programming
Node-RED	IoT dashboard and middleware
HiveMQ Broker	MQTT communication
Telegram Bot API	User notifications
Visual Studio Code / Thonny	Code development
GitHub	Version control and documentation

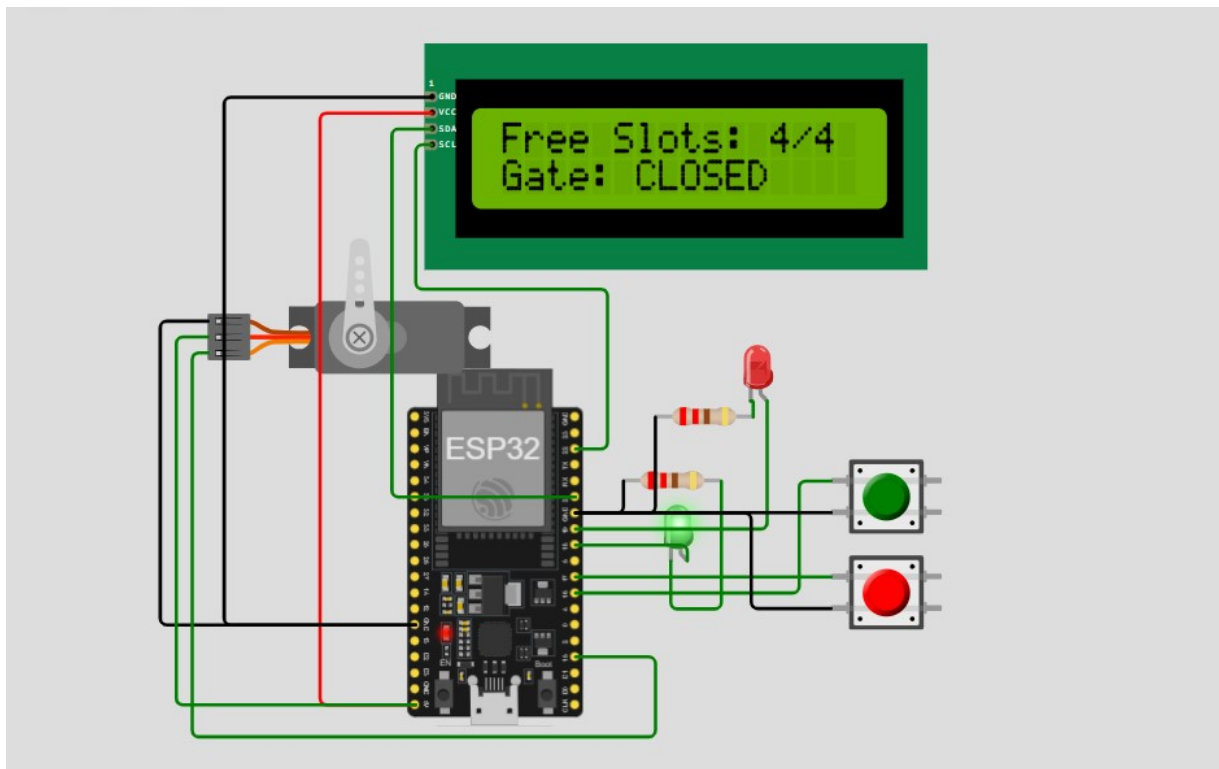


Fig1: Wokwi Simulation of the Smart Parking Management System

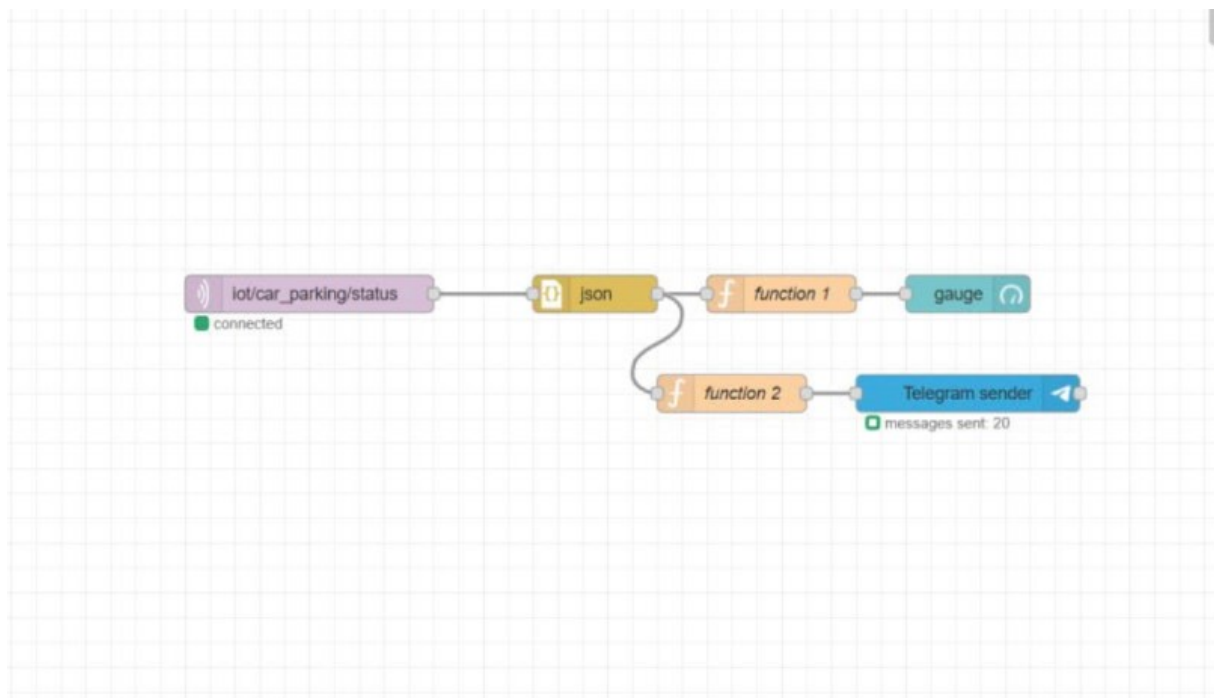


Fig2: Node-RED Flow Design.

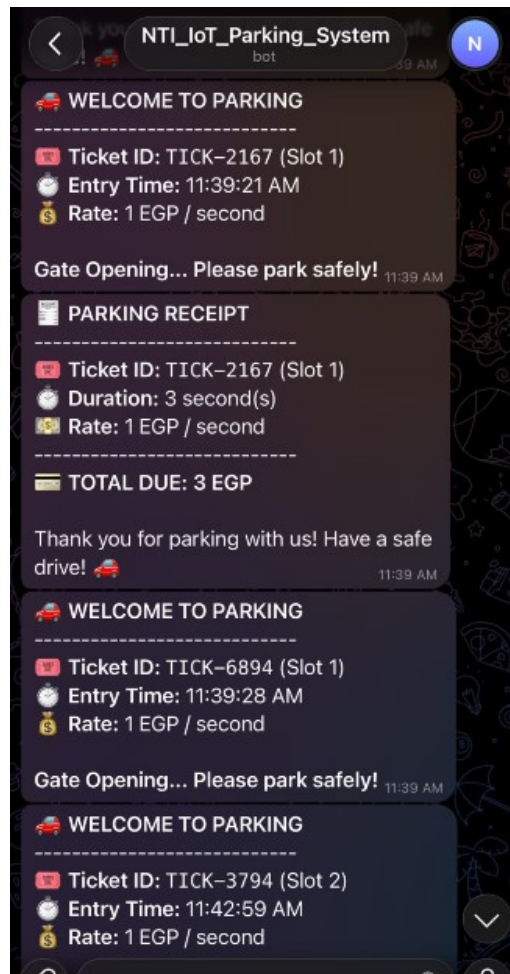


Fig4: Telegram Bot notifications showing the parking ticket and payment receipt.